

HW #3

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water

a) Given: $P = 100 \text{ lbf/in}^2$

$T = ?^\circ\text{F}$

$u = 500 \text{ BTU/lb}$

$v = ? \text{ ft}^3/\text{lb}$

$h = ? \text{ BTU/lb}$

$$u_{\text{mix}} = u_f + x(u_g - u_f)$$

$$x = \frac{u_{\text{mix}} - u_f}{u_g - u_f} = \frac{500 - 298.3}{1105.8 - 298.3}$$

$$\Rightarrow \boxed{x = 0.25}$$

$$\boxed{T = 327.86^\circ\text{F}}$$

$$v_{\text{mix}} = v_f + x(v_g - v_f)$$

$$v_f = 0.01774 \text{ ft}^3/\text{lb}$$

$$h_{\text{mix}} = h_f + x(h_g - h_f)$$

$$v_g = 4.434 \text{ ft}^3/\text{lb}$$

$$v_{\text{mix}} = 0.01774 \text{ ft}^3/\text{lb} + 0.25(4.434 - 0.01774)$$

$$\boxed{v_{\text{mix}} = 1.12 \text{ ft}^3/\text{lb}}$$

$$u_f = 298.3 \text{ BTU/lb}$$

$$u_g = 1105.8 \text{ BTU/lb}$$

$$h_{\text{mix}} = 298.6 \text{ BTU/lb} + 0.25(1187.8 - 298.6) \text{ BTU/lb}$$

$$h_f = 298.6 \text{ BTU/lb}$$

$$\boxed{h_{\text{mix}} = 521 \text{ BTU/lb}}$$

$$h_g = 1187.8 \text{ BTU/lb}$$

ammonia

B) Given: $T = 40^\circ\text{F}$; $x = 0.5$

$P = ? \text{ lbf/in}^2$

$$v_{\text{mix}} = v_f + x(v_g - v_f)$$

$v = ? \text{ ft}^3/\text{lb}$

$$= 0.02533 + 0.5(3.9664 - 0.02533)$$

$u = ? \text{ BTU/lb}$

$$\boxed{v = 1.996 \text{ ft}^3/\text{lb}}$$

$$\text{a) } T = 40^\circ\text{F}$$

$$u_{\text{mix}} = u_f + x(u_g - u_f)$$

$$\boxed{P = 73.359 \text{ lbf/in}^2}$$

$$= 85.98 + 0.5(568.42 - 85.98)$$

$$v_f = 0.02533 \text{ ft}^3/\text{lb}$$

$$\boxed{u = 327.2 \text{ BTU/lb}}$$

$$v_g = 3.9664 \text{ ft}^3/\text{lb}$$

$$u_f = 85.98 \text{ BTU/lb}$$

$$u_g = 568.42 \text{ BTU/lb}$$

water

c) Given: $P = 80 \text{ bar}$

$T = ?^\circ\text{C}$ $u = ? \text{ kJ/kg}$

$v = 0.04034 \text{ m}^3/\text{kg}$

$h = ? \text{ kJ/kg}$

a) $P = 80 \text{ bar}$ and $v = 0.04034 \text{ m}^3/\text{kg}$

Table A-4

$T = 480^\circ\text{C}$

$u = 3025.7 \text{ kJ/kg}$

$h = 3348.4 \text{ kJ/kg}$

d) Refrigerant 134a

Given: $T = -20^\circ\text{C}$

$P = ? \text{ bar}$

$h = h_g = 235.31 \text{ kJ/kg}$

$v = ? \text{ m}^3/\text{kg}$

$u = ? \text{ kJ/kg}$

a) $T = -20^\circ\text{C}$ and $h = 235.31 \text{ kJ/kg}$

Table A-10

$P = 1.3299 \text{ bar}$

$v = 0.1464 \text{ m}^3/\text{kg}$

$u = 215.84 \text{ kJ/kg}$

water

1 bar = 0.1 MPa

#2 Given: Rigid tank = constant volume

Sat Vapor $T_1 = 180^\circ\text{C}$ $P_1 = 7 \text{ MPa}$

$T_2 = 100^\circ\text{C}$ $x_2 = ?$

State 1 Sat Vapor

$T_1 = 180^\circ\text{C}$

at $T_1 = 180^\circ\text{C}$

Table A-2

$P_1 = 10.02 \text{ bar} \Rightarrow 1.002 \text{ MPa}$

$v_{\text{sat}} = 0.1941 \text{ m}^3/\text{kg}$

State 2

$$v_{1,\text{mix}} = v_{2,\text{mix}}$$

$T_2 = 100^\circ\text{C}$

$v_{\text{mix}} = 0.1941 \text{ m}^3/\text{kg}$

at $T_2 = 100^\circ\text{C}$

Table A-2

$P_2 = 1.014 \text{ bar} \Rightarrow 0.1014 \text{ MPa}$

$v_F = 1.0435 \times 10^{-3} \text{ m}^3/\text{kg}$

$v_g = 1.673 \text{ m}^3/\text{kg}$

$$v_{2,\text{mix}} = v_F + x(v_g - v_F)$$

$$x_2 = \frac{v_{\text{mix}} - v_{F_2}}{v_{g_2} - v_{F_2}} = \frac{0.1941 - 1.0435 \times 10^{-3}}{1.673 - 1.0435 \times 10^{-3}}$$

$x_2 = 0.115$

water

#3 Given: $P = \text{constant} = 300 \text{ kPa} (3 \text{ bar})$

$$m = 2 \text{ kg}$$

$$V_2 = 2.064 \text{ m}^3$$

$$v = \frac{V}{m} \Rightarrow v_2 = \frac{2.064 \text{ m}^3}{2 \text{ kg}} = 1.032 \text{ m}^3/\text{kg}$$

$$\boxed{v_2 = 1.032 \text{ m}^3/\text{kg}}$$

State 1

$$P_1 = 3 \text{ bar}$$

$$a) P_1 = 3 \text{ bar}$$

Table A-4

$$a) \boxed{T_1 = T_{\text{sat}} = 133.6^\circ \text{C}}$$

$$v_{\text{sat}} = 0.606 \text{ m}^3/\text{kg}$$

$$c) W = P_m (v_2 - v_1)$$

$$W = (300 \times 10^3 \text{ N/m}^2) (2 \text{ kg}) (1.032 - 0.606) \text{ m}^3/\text{kg}$$

$$W = 2556 \text{ N}\cdot\text{m}$$

$$\boxed{W = 2.556 \text{ kJ}}$$

State 2

$$P_2 = 3 \text{ bar}$$

$$v_2 = 1.032 \text{ m}^3/\text{kg}$$

$$a) P_2 = 3 \text{ bar and } v_2 = 1.032 \text{ m}^3/\text{kg}$$

Table A-4

$$b) \boxed{T_2 = 400^\circ \text{C}}$$

Refrigerant 134a

#4

$$m = 1 \text{ kg}$$

$$W = 20 \text{ kJ}$$

State 1

$$P_1 = 10 \text{ bar}$$

$$T_1 = 60^\circ \text{C}$$

Table A-12

$$\text{a) } P_1 = 10 \text{ bar and } T_1 = 60^\circ \text{C}$$

$$v_1 = 0.02301 \text{ m}^3/\text{kg}$$

$$u_1 = 268.35 \text{ kJ/kg}$$

State 2

$$P_2 = 2 \text{ bar}$$

$$T_2 = 30^\circ \text{C}$$

Table A-12

$$\text{a) } P_2 = 2 \text{ bar and } T_2 = 30^\circ \text{C}$$

$$u_2 = 253.06 \text{ kJ/kg}$$

Initial volume V_1

$$\text{a) } v = \frac{V}{m} \Rightarrow V_1 = v_1 \cdot m$$
$$= 0.02301 \text{ m}^3/\text{kg} (1 \text{ kg})$$

$$V_1 = 0.02301 \text{ m}^3$$

b) heat transfer Q

$$Q = W + m(u_2 - u_1)$$

$$= 20 \text{ kJ} + 1 \text{ kg} (253.06 - 268.35) \text{ kJ/kg}$$

$$Q = 4.71 \text{ kJ}$$

Water

#5 Given: Closed, rigid tank = constant volume

$$m = 1.5 \text{ kg}$$

$$T_1 = 70^\circ \text{C}$$

$$T_2 = 120^\circ \text{C}$$

State 1 - liquid vapor

$$T_1 = 70^\circ \text{C}$$

$$P_1 = 0.3119 \text{ bar} \Rightarrow 31.19 \text{ kPa}$$

$$v_f = 1.0228 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$v_g = 5.042 \text{ m}^3/\text{kg}$$

$$u_f = 292.95 \text{ kJ/kg}$$

$$u_g = 2469.6 \text{ kJ/kg}$$

$$v_{\text{mix}} = v_f + x(v_g - v_f)$$

$$x = \frac{v_{\text{mix}} - v_f}{v_g - v_f} = \frac{0.8919 - 1.0228 \times 10^{-3}}{5.042 - 1.0228 \times 10^{-3}} = 0.177$$

State 2 - Sat vapor

$$v_2 = v_{\text{mix}} = 0.8919 \text{ m}^3/\text{kg}$$

$$T_2 = 120^\circ \text{C}$$

$$v_g = v_{\text{mix}} = 0.8919 \text{ m}^3/\text{kg}$$

$$u_2 = u_g = 2829.3 \text{ kJ/kg}$$

$$u_{\text{mix}} = u_f + x(u_g - u_f)$$

$$= 292.95 + 0.177(2469.6 - 292.95)$$

$$u_{\text{mix}} = 678.22 \text{ kJ/kg}$$

$$Q = m \Delta u$$

$$Q = (2829.3 - 678.22) \text{ kJ/kg} (1.5 \text{ kg})$$

$$Q = 2776.62 \text{ kJ}$$

Refrigerant 22

#6

Given: $P = 10 \text{ bar}$

$V = 7 \text{ m}^3$

$m_f = 25 \text{ kg}$

① $x = 60\% = 0.6$

② $P = 10 \text{ bar}$

table A-8

② $v_f = 0.8352 \times 10^{-3} \text{ m}^3/\text{kg}$

$v_g = 0.0236 \text{ m}^3/\text{kg}$

$$v_{\text{mix}} = v_f + x(v_g - v_f)$$

$$= 0.8352 \times 10^{-3} \text{ m}^3/\text{kg} + 0.6(0.0236 - 0.8352 \times 10^{-3}) \text{ m}^3/\text{kg}$$

$v_{\text{mix}} = 0.01449 \text{ m}^3/\text{kg}$

③ $x = \frac{\text{mass Sat Vapor}}{\text{total mass}} = \frac{m_g}{m_f + m_g}$

$$x = \frac{m_g}{m_f + m_g}$$

⑤ $V_f = v_f \cdot m_f$
 $= 0.8352 \times 10^{-3} \text{ m}^3/\text{kg} (25 \text{ kg})$

$$0.6 = \frac{m_g}{m_g + 25}$$

$V_f = 0.02088 \text{ m}^3$

volume of tank

$$(m_g + 25)0.6 = m_g$$

$$V = 0.01449 \text{ m}^3/\text{kg} (25 + 37.5) \text{ kg}$$

$$0.6m_g + 15 = m_g$$

$V = 0.9056 \text{ m}^3$

$$-0.4m_g$$

$$0.4m_g = 15$$

$m_g = 37.5 \text{ kg}$

⑥ $x_v = \frac{V_g}{V_{\text{tank}}} \times 100$

$$= \frac{0.885}{0.9056} \times 100 = 97.725\%$$

④

$$v_g = \frac{V_g}{m} \Rightarrow V_g = v_g \cdot m_g$$

$$V_g = 0.0236 \text{ m}^3/\text{kg} (37.5 \text{ kg})$$

$V_g = 0.885 \text{ m}^3$

$$1g = 10^{-3}kg$$

Helium

#7 Given: $m = 0.2g \Rightarrow 2 \times 10^{-4}kg$

$$P_1 = 1.3 \text{ bar} \Rightarrow 0.13 \text{ MPa}$$

$$P_2 = 0.80 \text{ bar} \Rightarrow 0.08 \text{ MPa}$$

(1) ~~$T_1 = 27^\circ\text{C}$~~ $T_1 = 27^\circ\text{C}$

$$T_2 = 17^\circ\text{C}$$

$$T(^{\circ}\text{K}) = 27 + 273.15$$

(2) $T(^{\circ}\text{K}) = T(^{\circ}\text{C}) + 273.15$

$$= 17 + 273.15$$

$$T_1 = 300.15 \text{ K}$$

$$T_2 = 290.15 \text{ K}$$

(3) $\bar{R} = 8.314 \text{ kJ/kmol} \cdot \text{K}$

$$R = \frac{\bar{R}}{M}$$

Helium Table A-1

$$M = 4.003 \text{ kg/kmol}$$

$$R = \frac{8.314 \text{ kJ/kmol}}{4.003 \text{ kg/kmol}} = \boxed{2.077} \text{ kJ/kg} \cdot \text{K}$$

(4) $P_1 V_1 = m R T_1$

$$V_1 = \frac{m R T_1}{P_1} = \frac{2 \times 10^{-4} \text{ kg} (2.077 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}) (300.15 \text{ K})}{130 \text{ kPa}}$$

$$V_1 = 9.591 \times 10^{-4} \text{ m}^3$$

$$\frac{9.591 \times 10^{-4} \text{ m}^3}{1 \text{ m}^3} \times \frac{10^6 \text{ cm}^3}{1 \text{ m}^3}$$

$$\boxed{V_1 = 959.1 \text{ cm}^3}$$

$$V_2 = \frac{m R T_2}{P_2} = \frac{2 \times 10^{-4} (2.077 \text{ kJ/kg} \cdot \text{K}) (290.15 \text{ K})}{80 \text{ kPa}}$$

$$V_2 = 1.507 \times 10^{-3} \text{ m}^3$$

$$\boxed{V_2 = 1506.6 \text{ cm}^3}$$

CO₂

#8

$$V = \text{Constant} = 0.05 \text{ m}^3$$

$$\text{Given: } P_1 = 2.75 \text{ bar} = 275 \text{ kPa}$$

$$T_1 = 25^\circ \text{C}$$

$$T_2 = 257^\circ \text{C}$$

$$Q = -2.75 \text{ kJ}$$

$$R = \frac{\bar{R}}{M} = \frac{8.314 \text{ kJ/kmol} \cdot \text{K}}{44.01 \text{ kg/kmol}} = 0.1889 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$\bar{R} = 8.314 \text{ kJ/kmol} \cdot \text{K}$$

$$M_{\text{CO}_2} = 44.01 \text{ kg/kmol}$$

$$T_1 = 25^\circ \text{C}$$

$$T_2 = 257^\circ \text{C}$$

$$T(^{\circ}\text{K}) = T(^{\circ}\text{C}) + 273.15$$
$$= 25 + 273.15$$

$$T(^{\circ}\text{K}) = 257 + 273.15$$

$$T_2 = 530 \text{ K}$$

$$T_1 = 298 \text{ K}$$

$$\bar{u}_2 = 14,622 \text{ kJ/kmol}$$

$$\bar{u}_1 = 6,885 \text{ kJ/kmol}$$

$$m = \frac{P_1 V}{RT_1} = \frac{275 \text{ kPa} (0.05 \text{ m}^3)}{0.1889 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} (298 \text{ K})}$$

$$m = 0.244261 \text{ kg}$$

$$Q = W + \Delta U$$

$$-2.75 \text{ kJ} = W + \frac{m}{M} (\bar{u}_2 - \bar{u}_1)$$

$$-2.75 \text{ kJ} = W + \frac{0.244261 \text{ kg}}{44.01 \text{ kg/kmol}} [14,622 - 6,885] \text{ kJ/kmol}$$

$$W = -45.69 \text{ kJ}$$